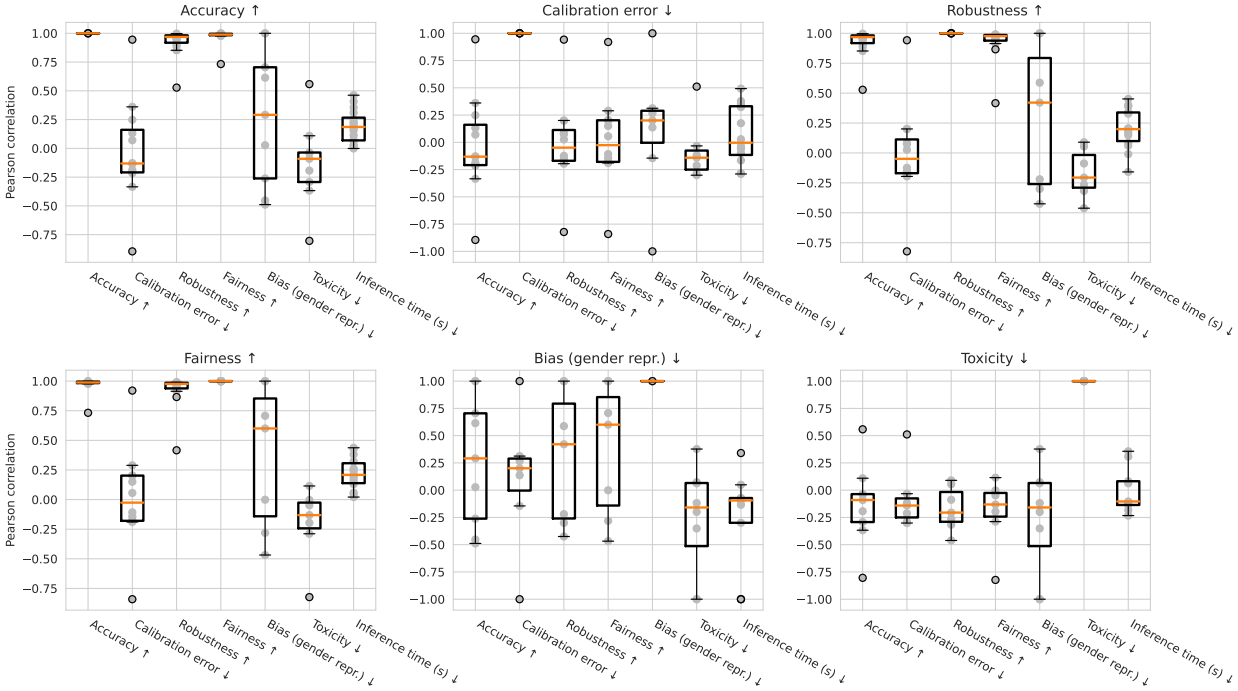

Figure 25: **Correlation between metrics.** The Pearson correlation between each metric and every other metric (x-axis). The small grey dots denote the correlation on each individual scenario. Trends are qualitatively similarly for other correlation measures (e.g. Spearman correlation). For calibration error, we measure ECE-10; for bias, we measure bias in gender representation; and for efficiency, we measure denoised inference time.

Shifting focus to calibration, the trends are more scenario-dependent, as can be seen for accuracy by the visual clustering of points for a given scenario in Figure 24. Strikingly, we find that the relationship between accuracy and calibration can be quite different for very similar scenarios: for two commonsense-centric QA scenarios, we see accuracy and calibration are highly correlated in **OpenBookQA** (correlation of greater than 0.8) whereas accuracy and calibration **error** are highly correlated in **HellaSwag** (correlation greater than 0.85). Further, while not unsurprising given the strong correlations between accuracy, robustness, and fairness, the finding that more robust and more fair models can be less well-calibrated is counter-intuitive and surprising.

For generative harms, we also see significant variation in the relationship with other metrics. For toxicity, the correlations on average with other metrics are near zero as the toxicity rates themselves for toxicity are relatively constant and near zero, with the clear exception of **NarrativeQA** in Figure 24. In contrast for bias, we see some positive correlation between accuracy and gender representation bias (lower is better), with an even more striking average correlation of scenarios of more the 0.5 between fairness (higher is better) and gender representation bias (lower is better). In other words, we see a clear and surprising trade-off: models that tend to have better fairness performance, which depends on task-specific performance, tend to have worse gender bias, which depends on model generations but not task-specific performance. As to the two generative harms, we see there are some scenarios where they are fairly anti-correlated, which indicates the less gender biased models tend to be more toxic for these scenarios. These findings indicate it may not suffice to only measure one of these types of harms, and that efforts to reduce one may have side-effects for the other (cf. Xu et al., 2021), which is especially relevant given the harms of toxicity are more straightforward/immediate (e.g. more likely to draw media/journalistic attention) whereas the harms of bias may be more subtle/gradual.

Finally, for efficiency, we see fairly weak correlations with other metrics. In particular, we do not see a strong overarching trade-off between accuracy and efficiency, which we may have anticipated *a priori*. This is made clear by Figure 24, which instead showcases that the trends for efficiency tend to be quite scenario-dependent. This is especially true for the denoised metric, given a lot of what drives the measurements relates to the fundamental distributional properties of the scenario (i.e. the length of inputs and outputs).

Direct model comparisons. A central objective for our evaluation is to achieve common and unified understanding of all the models available. In particular, as we document in Appendix F, the 30 models we evaluate in this work have been previously evaluated under different conditions for different evaluation suites. In particular, the degree of overlap is surprisingly low and uneven across different models. Consequently, we lack clarity as a field on the relationship between different models.

In Figure 26, we report how different models would fare in a head-to-head comparison for each metric across all the core scenarios.[64] We find that text-davinci-002 is clearly the most accurate model, winning in these comparisons more than 90% of the time. Of the remaining models, we see that the largest model TNLG v2 (530B) is the second most accurate, followed by Anthropic-LM v4-s3 (52B). In light of the significant difference in model scale of a factor of more than 10 between TNLG v2 (530B) and Anthropic-LM v4-s3 (52B), given text-davinci-002 and Anthropic-LM v4-s3 (52B) share instruction-tuning in common (which other models, beyond smaller OpenAI text variants, do not), this suggests that instruction-tuning and the use of human feedback is an efficient and effective means for improving model accuracy.

Further, while we will later see the overall relationship between model scale and accuracy can be complicated (Figure 29), here we see a clear thresholding effect in terms of model scale. All of the models with a win rate for accuracy clearly above chance (e.g. 55% or more) have more than 50B parameters, whereas the sole exception of a large model outside this group of the top 10 is YaLM (100B). YaLM (100B) has a surprisingly poor accuracy win rate below 25%, perhaps because of significant training on Russian instead of English). Within the top 10, we see model scale appears to be less predictive of rank, as two of the largest models (BLOOM (176B) and J1-Jumbo v1 (178B); both 100B+ parameters) are at the bottom of this tier, whereas Anthropic-LM v4-s3 (52B) and Cohere xlarge v20220609 (52.4B) (the two smallest models in the tier) are

[64] We exclude efficiency as we believe head-to-head comparisons based solely on efficiency are not meaningful; instead efficiency needs to be considered in conjunction with other metrics like accuracy. Forthcoming work (Narayanan et al., Forthcoming) examines the tradeoffs between accuracy and efficiency.

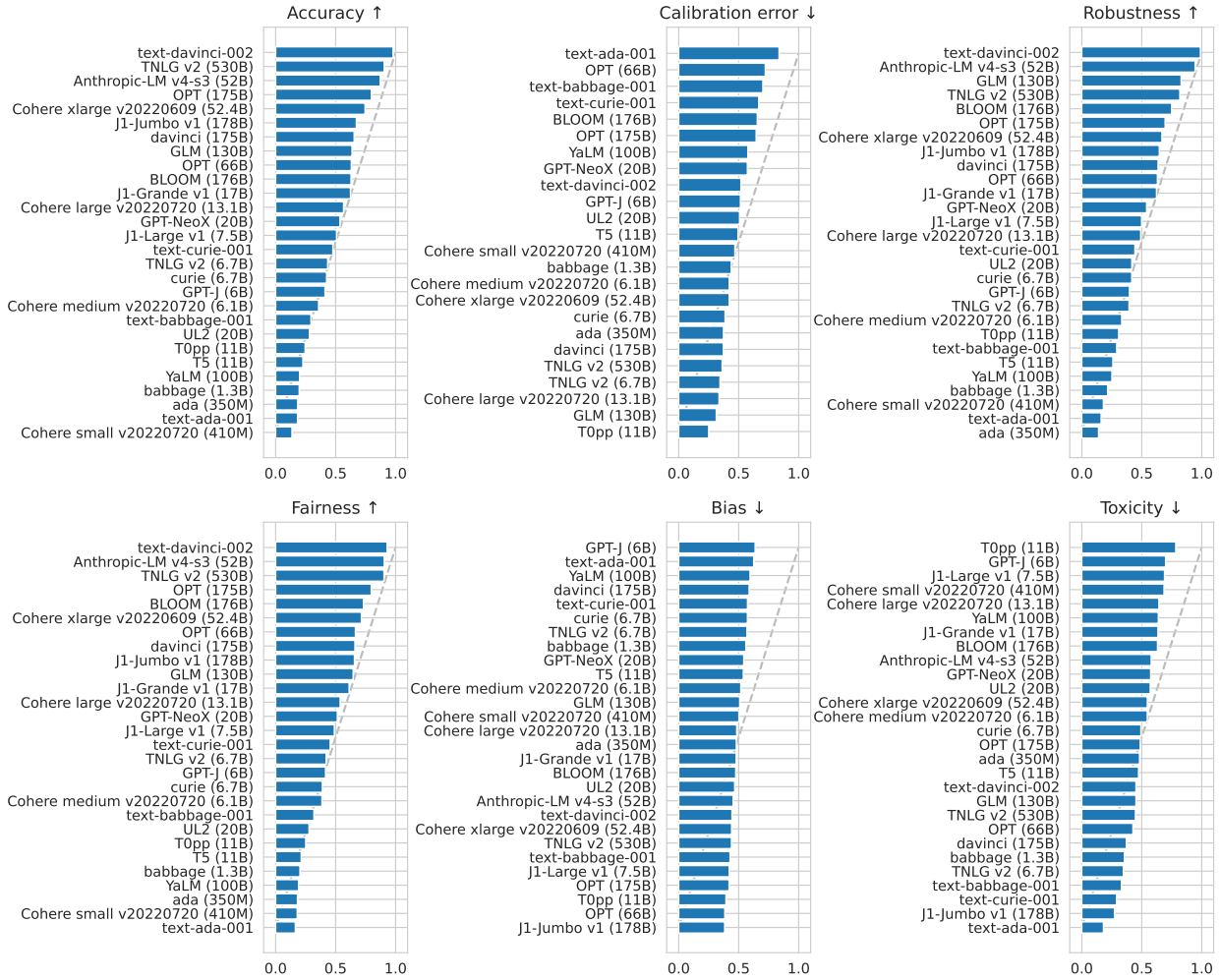

Figure 26: **Head-to-head win rate per each model.** We report the fraction of head-to-head comparisons between the given model and all other models, across all scenarios, where the given model is higher along the metric (e.g. more accurate in the accuracy subfigure). If a model was the highest for the given metric for every scenario, it would receive a score of 1.0; if a model received a score of 0.5, then if a scenario and second model were chosen at random, the outcome of the comparison would be a coin flip. For metrics where higher is better (Accuracy, Robustness, Fairness), the better models are towards the top, and vice versa. For calibration error, we measure ECE-10; for bias, we measure bias in gender representation; and for efficiency, we believe these comparisons should not be made without consideration of accuracy and they are further discussed in Narayanan et al. (Forthcoming).